UNIT-1

- Define consensus mechanism
- Define block in blockchain technology.
- Classify the structure of blocks in block-chain technology.
- Define public ledger in the context of block-chain.
- Enlist two benefits of using block-chain technology.
- Classify ledger and block in blockchain technology.
- Give examples of block-chain technology in everyday life
- Enlist four applications of block-chain technology
- Classify the structure of blocks in block-chain technology.
- Give examples of block-chain technology in everyday life
- Enlist two benefits of using block-chain technology.
- Write short notes on miners and also describe the role of miners in a blockchain network
- Write short notes on distributed and centralized systems in the blockchain network.
- Illustrate the complete transaction process in block-chain, covering the following key components:
    - Transaction
    - Consensus
    - The Chain
    - The Longest Chain
- Write short note on:
  a) Distributed Ledger Technology
  b) Blockchain 2.0
- Write the different types of block-chain and explain each of them.
- Describe the working of block-chain with the help of a diagram.
- Discuss the role of smart contracts in Block-chain 2.0.

UNIT-2

- State any two cryptographic primitives used in block-chain.
- List out the examples which belong to permission-ed blockchain technology.
- Restate the term Merkle Tree in block-chain technology.
- State any two properties of a good hash function
- Distinguish between public and private block-chains (any four points).
- Illustrate how hashing ensures data integrity in block-chain.
- Explain cryptographic concepts in blockchain technology.
- State any four challenges of block-chain technology.
- List out the types of fork in block-chain
- Restate the concept of bitcoin technology.
- State the term message digest
- State any two cryptographic primitives used in block-chain.
- List out the examples which belong to permission-ed blockchain technology.
- Restate the concept of bitcoin technology.

- State the term message digest.
- Illustrate the types of fork in block-chain.
- Explain cryptographic concepts in blockchain technology.
- Distinguish between permission-ed and permission-less block-chain technology.
- Illustrate the working of public key cryptosystems in block-chain technology.
- Explain the layered architecture of block-chain technology.

UNIT-3

- Write the concept of a mining pool.
- Define public ledger in the context of block-chain
- Give examples of digital currency in different technologies.
- Discuss digital signatures and their properties.
- Write the role of miners in the Bitcoin block-chain
- Define Hashing Mechanism.
- Discuss double spending terms in block-chain or bitcoin.
- Analyze the Byzantine algorithm and explain it in block-chain technology..
- Analyze the PAXOS algorithm and explain it in blockchain technology
- Rewrite about following:

a) Proof of Work
b) Proof of Stake
c) Proof of Burn
d) Proof of Elapsed Time
e) Proof of Capacity
f) Proof of Authority

- Rewrite about following:

a) Mining Process
b) Miner's role
c) Mining Pool
d) Mining Difficulty

UNIT-4

- Identify the inventor and currency of ethernet
- Explain EVM in Ethereum Technology
- Identify the term centralized in Ethereum
- Examine the working of ethereum technology
- Examine wallets,accounts,keys and addresses in ethereum
- Examine the smart contract concept in Ethereum Technology
- Explain different(any six) operators and control statements in solidity programming

UNIT-5

- Rewrite the concept of hyper-ledger technology.
- Sketch different nodes in hyperledger fabric.

- Rewrite the advantages of hyper-ledger technology.
- Discuss types of channels in hyper-ledger fabric.
- Sketch the concept of smart contract in hyper-ledger fabric.
- Describe centralized and decentralized network technology in hyper-ledger fabric.
- Describe hyperledger fabric, history, features ,application,nodes and working of it.
- Describe transactions,consensus protocol ,nodes in hyperledger fabric.